HW6

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Q1: 2.28

Let μ be the Lebesgue measure on $([-1,1], \mathcal{B}([-1,1]))$. For $n \geq 1$, define $f_n(x) = nI_{(0,n^{-1})}(x) - nI_{(-n^{-1},0)}(x)$ and $f(x) \equiv 0$ for $x \in [-1,1]$. Show that $f_n \to f$ a.e. (μ) and $\int f_n \, d\mu \to \int f \, d\mu$ but $\{f_n\}_{n\geq 1}$ is not UI.

Answer

We utilize the following results:

- Proposition A.2.1 (Axiom of Choice): for every $y \in \mathbb{R}$, there exists $n \in \mathbb{N}$ with $n > y$.
- Definition 2.3.2 (integral of a simple nonnegative function): if $f = \sum_{i=1}^k c_i I_{A_i}$ with $c_i \geq 0$ and $A_i \in \mathcal{F}$, then $\int f \, d\mu = \sum_{i=1}^k c_i \, \mu(A_i)$.
- Definition 2.3.4 (integral of a measurable function): $\int f \, d\mu = \int f^+ \, d\mu - \int f^- \, d\mu$, where $f^+ = fI_{\{f\geq 0\}}$ and $f^- = -fI_{\{f<0\}}$, provided at least one of the two is finite.
- Definition 2.5.2 (convergence a.e.): $f_n \to f$ a.e. (μ) if there is a $B \in \mathcal{F}$ with $\mu(B) = 0$ and $f_n(\omega) \to f(\omega)$ for all $\omega \in B^c$.
- Definition 2.5.8 (uniform integrability): $\{f_\lambda : \lambda \in \Lambda\} \subset L^1(\Omega, \mathcal{F}, \mu)$ is UI if $\sup_{\lambda\in\Lambda} a_\lambda(t) \to 0$ as $t \to \infty$, where $a_\lambda(t) \equiv \int_{\{|f_\lambda|>t\}} |f_\lambda| \, d\mu$.
- Remark 2.5.4: when $\mu(\Omega) < \infty$ and $f_n \to f$ a.e. (μ), UI of $\{f_n\}_{n\geq 1}$ is sufficient, but not necessary, for $\int f_n \, d\mu \to \int f \, d\mu$; when $\mu(\Omega) = \infty$, UI no longer suffices.

Proof. Since f_n equals n on $(0, n^{-1})$, $-n$ on $(-n^{-1}, 0)$, and 0 elsewhere,

$$f_n^+ = nI_{(0,n^{-1})}, \qquad f_n^- = nI_{(-n^{-1},0)}, \qquad |f_n| = nI_{(-n^{-1},0)\cup(0,n^{-1})}.$$

$f_n \to f$ a.e. (μ): Let $x \in [-1,1]$ be given. If $x = 0$, then $f_n(0) = 0$ for all $n \geq 1$, since 0 is in neither open interval. If $x \neq 0$, then by Proposition A.2.1 there exists $N \in \mathbb{N}$ with $N > 1/|x|$. For all $n \geq N$, $1/n \leq 1/N < |x|$, so $x \notin (-n^{-1}, n^{-1})$ and $f_n(x) = 0$. In either case, $f_n(x) \to 0 = f(x)$. Thus $B \equiv \{x \in [-1,1] : f_n(x) \not\to f(x)\} = \emptyset$, and since $\mu(\emptyset) = 0$, $f_n \to f$ a.e. (μ) (Definition 2.5.2).

$\int f_n \, d\mu \to \int f \, d\mu$: By Definitions 2.3.2 and 2.3.4, for every $n \geq 1$,

$$\int f_n \, d\mu = \int f_n^+ \, d\mu - \int f_n^- \, d\mu = n\,\mu\big((0,n^{-1})\big) - n\,\mu\big((-n^{-1},0)\big) = 1 - 1 = 0.$$

Since $\int f \, d\mu = \int 0 \, d\mu = 0$, we have $\int f_n \, d\mu = \int f \, d\mu$ for every n, and so $\int f_n \, d\mu \to \int f \, d\mu$.

$\{f_n\}_{n\geq 1}$ is not UI: By Definition 2.3.2, $\int |f_n| \, d\mu = n \cdot \frac{2}{n} = 2 < \infty$, so each $f_n \in L^1(\mu)$. Let $t > 0$ be given. If $n \leq t$, then $|f_n| \leq n \leq t$, so $\{|f_n| > t\} = \emptyset$ and $a_n(t) = 0$. If $n > t$, then $\{|f_n| > t\} = (-n^{-1}, 0) \cup (0, n^{-1})$, so $a_n(t) = \int |f_n| \, d\mu = 2$. It then follows that

$$\sup_{n\geq 1} a_n(t) = 2 \quad \text{for every } t > 0,$$

which does not tend to 0 as $t \to \infty$. Thus $\{f_n\}_{n\geq 1}$ is not UI.

Said differently, even with $\mu(\Omega) < \infty$, UI is not necessary for $\int f_n \, d\mu \to \int f \, d\mu$ (see Remark 2.5.4). //

Q2: 2.30

For $n \geq 1$, let $f_n(x) = n^{-1/2} I_{(0,n)}(x)$, $x \in \mathbb{R}$, and let $f(x) = 0$, $x \in \mathbb{R}$. Let m denote the Lebesgue measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. Show that $f_n \to f$ a.e. (m) and $\{f_n\}_{n\geq 1}$ is UI, but $\int f_n \, dm \not\to \int f \, dm$.

Answer

We utilize the following results (each stated in Q1):

- Definition 2.3.2 (integral of a simple nonnegative function).
- Definition 2.5.2 (convergence a.e.).
- Definition 2.5.8 (uniform integrability), with $a_n(t) \equiv \int_{\{|f_n|>t\}} |f_n| \, dm$.
- Remark 2.5.4: when $\mu(\Omega) < \infty$ and $f_n \to f$ a.e. (μ), UI of $\{f_n\}_{n\geq 1}$ is sufficient, but not necessary, for $\int f_n \, d\mu \to \int f \, d\mu$; when $\mu(\Omega) = \infty$, UI no longer suffices.

Proof.

$f_n \to f$ a.e. (m): For every $x \in \mathbb{R}$, $0 \leq f_n(x) \leq n^{-1/2}$, and $n^{-1/2} \to 0$. So $f_n(x) \to 0 = f(x)$ for every $x \in \mathbb{R}$. Thus $B \equiv \{x : f_n(x) \not\to f(x)\} = \emptyset$, and since $m(\emptyset) = 0$, $f_n \to f$ a.e. (m) (Definition 2.5.2).

$\{f_n\}_{n\geq 1}$ is UI: By Definition 2.3.2, $\int |f_n| \, dm = n^{-1/2} \, m((0,n)) = n^{-1/2} \cdot n = n^{1/2} < \infty$, so each $f_n \in L^1(m)$. Let $t \geq 1$ be given. Since $|f_n| \leq n^{-1/2} \leq 1 \leq t$, $\{|f_n| > t\} = \emptyset$, and so $a_n(t) = 0$ for all $n \geq 1$. Hence $\sup_{n\geq 1} a_n(t) = 0$ for all $t \geq 1$, which tends to 0 as $t \to \infty$. Thus $\{f_n\}_{n\geq 1}$ is UI.

$\int f_n \, dm \not\to \int f \, dm$: From the computation above, $\int f_n \, dm = n^{1/2} \to \infty$, while $\int f \, dm = 0$. Thus $\int f_n \, dm \not\to \int f \, dm$.//

Q3: 2.36

We utilize the following results for (a) and (b):

- Definition 2.5.3 (convergence in measure): $f_n \xrightarrow{m} f$ if $\mu(\{|f_n - f| > \epsilon\}) \to 0$ for each $\epsilon > 0$.
- Definition 2.5.4 (convergence in L^p): $f_n \to f$ in $L^p(\mu)$ if $f_n, f \in L^p(\mu)$ and $\int |f_n - f|^p \, d\mu \to 0$.
- Definition 2.5.8 (uniform integrability; see Q1), with $a_\lambda(t) \equiv \int_{\{|f_\lambda|>t\}} |f_\lambda| \, d\mu$.
- Proposition 2.5.7 (properties of UI families of integrable functions): (i) a finite family is UI; (iv) if $\{f_\lambda : \lambda \in \Lambda\}$ and $\{g_\gamma : \gamma \in \Gamma\}$ are UI, then so is $\{f_\lambda + g_\gamma : \lambda \in \Lambda, \gamma \in \Gamma\}$; (v) if $\{f_\lambda : \lambda \in \Lambda\}$ is UI and $\mu(\Omega) < \infty$, then $\sup_{\lambda\in\Lambda} \int |f_\lambda| \, d\mu < \infty$.
- Theorem 2.5.2: if $f_n \xrightarrow{m} f$, then there is a subsequence $\{n_k\}_{k\geq 1}$ with $f_{n_k} \to f$ a.e. (μ).
- Theorem 2.3.7 (Fatou's lemma): for nonnegative measurable $\{f_n\}_{n\geq 1}$, $\int \liminf_{n\to\infty} f_n \, d\mu \leq \liminf_{n\to\infty} \int f_n \, d\mu$.
- Corollary 2.3.3: Proposition 2.3.1 holds for nonnegative measurable functions; in particular, $f \geq g$ a.e. (μ) implies $\int f \, d\mu \geq \int g \, d\mu$, and $f = g$ a.e. (μ) implies $\int f \, d\mu = \int g \, d\mu$.

(a)

Let $\{f_n\}_{n\geq 1} \subset L^1(\Omega, \mathcal{F}, \mu)$ such that $f_n \to 0$ in $L^1(\mu)$. Show that $\{f_n\}_{n\geq 1}$ is UI.

Answer

Proof. Let $a_n(t)$ be as in Q1, and let $\epsilon > 0$ be given. Since $f_n \to 0$ in $L^1(\mu)$, $\int |f_n| \, d\mu \to 0$ (Definition 2.5.4). So there exists $N \in \mathbb{N}$ with $\int |f_n| \, d\mu < \epsilon$ for all $n > N$. Since $a_n(t) \leq \int |f_n| \, d\mu$ for every t,

$$\sup_{n>N} a_n(t) \leq \epsilon \quad \text{for all } t > 0.$$

That leaves the first N terms. $\{f_1, \ldots, f_N\}$ is a finite family in $L^1(\mu)$, so it is UI by Proposition 2.5.7(i). Hence there exists $t_\epsilon > 0$ with $\sup_{1 \le n \le N} a_n(t) < \epsilon$ for all $t \ge t_\epsilon$. Combining the two bounds,

$$\sup_{n \ge 1} a_n(t) \le \epsilon \quad \text{for all } t \ge t_\epsilon.$$

Since $\epsilon > 0$ was arbitrary, $\sup_{n \ge 1} a_n(t) \to 0$ as $t \to \infty$, and we conclude that $\{f_n\}_{n \ge 1}$ is UI. //

(b)

Let $\{f_n\}_{n \ge 1} \subset L^p(\Omega, \mathcal{F}, \mu)$, $0 < p < \infty$, such that $\mu(\Omega) < \infty$, $\{|f_n|^p\}_{n \ge 1}$ is UI and $f_n \xrightarrow{m} f$. Show that $f \in L^p(\mu)$ and $f_n \to f$ in $L^p(\mu)$.

Answer

Proof. Write $g_n \equiv |f_n - f|^p$. We then have three goals:

- (1) $f \in L^p(\mu)$,
- (2) $\{g_n\}_{n \ge 1}$ is UI, and
- (3) $\int g_n \, d\mu \to 0$.

(1) $f \in L^p(\mu)$: By Theorem 2.5.2, there is a subsequence $\{n_k\}_{k \ge 1}$ with $f_{n_k} \to f$ a.e. (μ). Since $x \mapsto |x|^p$ is continuous, $|f_{n_k}|^p \to |f|^p$ a.e. (μ) as well, so $|f|^p = \liminf_{k \to \infty} |f_{n_k}|^p$ a.e. (μ). Since $\{|f_n|^p\}_{n \ge 1}$ is UI and $\mu(\Omega) < \infty$, Proposition 2.5.7(v) gives

$$K \equiv \sup_{n \ge 1} \int |f_n|^p \, d\mu < \infty.$$

It then follows from Corollary 2.3.3 and Fatou's lemma that

$$\int |f|^p \, d\mu = \int \liminf_{k \to \infty} |f_{n_k}|^p \, d\mu \le \liminf_{k \to \infty} \int |f_{n_k}|^p \, d\mu \le K < \infty,$$

so $f \in L^p(\mu)$.

(2) $\{g_n\}_{n \ge 1}$ is UI: Since $|f_n - f| \le |f_n| + |f| \le 2 \max\{|f_n|, |f|\}$ and $x \mapsto x^p$ is increasing on $[0, \infty)$,

$$0 \le g_n \le 2^p \max\{|f_n|^p, |f|^p\} \le G_n \equiv 2^p |f_n|^p + 2^p |f|^p.$$

We claim $\{G_n\}_{n \ge 1}$ is UI. First, $\{2^p |f_n|^p\}_{n \ge 1}$ is UI, since for $t > 0$,

$$\sup_{n \ge 1} \int_{\{2^p |f_n|^p > t\}} 2^p |f_n|^p \, d\mu = 2^p \sup_{n \ge 1} \int_{\{|f_n|^p > t/2^p\}} |f_n|^p \, d\mu \to 0 \quad \text{as } t \to \infty$$

by the UI of $\{|f_n|^p\}_{n \ge 1}$. Second, $2^p |f|^p \in L^1(\mu)$ by (1), so the one-element family $\{2^p |f|^p\}$ is UI by Proposition 2.5.7(i). Hence $\{G_n\}_{n \ge 1}$ is UI by Proposition 2.5.7(iv).

To pass this to $\{g_n\}_{n \ge 1}$, let $h_t(x) \equiv x I_{(t, \infty)}(x)$, so that $\int_{\{u > t\}} u \, d\mu = \int h_t(u) \, d\mu$ for nonnegative u. Since h_t is nondecreasing on $[0, \infty)$ and $0 \le g_n \le G_n$, we have $h_t(g_n) \le h_t(G_n)$, and so by Corollary 2.3.3,

$$\sup_{n \ge 1} \int_{\{g_n > t\}} g_n \, d\mu \le \sup_{n \ge 1} \int_{\{G_n > t\}} G_n \, d\mu \to 0 \quad \text{as } t \to \infty.$$

Each g_n is also integrable, since $g_n \le G_n \in L^1(\mu)$. Thus $\{g_n\}_{n \ge 1}$ is UI.

(3) $\int g_n \, d\mu \to 0$: Let $\epsilon > 0$ and $t > \epsilon$ be given, and let $c(t) \equiv \sup_{k \ge 1} \int_{\{g_k > t\}} g_k \, d\mu$. Splitting Ω into $\{g_n \le \epsilon\}$, $\{\epsilon < g_n \le t\}$, and $\{g_n > t\}$,

$$\int g_n \, d\mu \le \epsilon \, \mu(\Omega) + t \, \mu(\{g_n > \epsilon\}) + c(t).$$

Since $\{g_n > \epsilon\} = \{|f_n - f| > \epsilon^{1/p}\}$ and $f_n \xrightarrow{m} f$, Definition 2.5.3 gives $\mu(\{g_n > \epsilon\}) \to 0$. So

$$0 \le \limsup_{n\to\infty} \int g_n \, d\mu \le \epsilon \, \mu(\Omega) + c(t).$$

Letting $t \to \infty$, $c(t) \to 0$ by (2). Then letting $\epsilon \downarrow 0$, $\epsilon \, \mu(\Omega) \to 0$ since $\mu(\Omega) < \infty$. Thus $\int |f_n - f|^p \, d\mu \to 0$.

Combining (1)–(3), $f_n, f \in L^p(\mu)$ and $\int |f_n - f|^p \, d\mu \to 0$, so $f_n \to f$ in $L^p(\mu)$ (Definition 2.5.4). //

Q4: 2.42

(Change of variables formula). Let $(\Omega_i, \mathcal{F}_i)$, $i = 1, 2$ be two measurable spaces. Let $f : \Omega_1 \to \Omega_2$ be $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$-measurable, $h : \Omega_2 \to \mathbb{R}$ be $\langle \mathcal{F}_2, \mathcal{B}(\mathbb{R}) \rangle$-measurable, and μ_1 be a measure on $(\Omega_1, \mathcal{F}_1)$. Show that $g \equiv h \circ f$, i.e., $g(\omega) \equiv h(f(\omega))$ for $\omega \in \Omega_1$ is in $L^1(\mu)$ iff $h(\cdot) \in L^1(\Omega_2, \mathcal{F}_2, \mu_2)$ where $\mu_2 = \mu_1 f^{-1}$ iff $I(\cdot) \in L^1\big(\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_3 \equiv \mu_2 h^{-1}\big)$ where $I(\cdot)$ is the identity function in $\mathbb{R}$, i.e., $I(x) \equiv x$ for all $x \in \mathbb{R}$ and also that

$$\int_{\Omega_1} g \, d\mu_1 = \int_{\Omega_2} h \, d\mu_2 = \int_{\mathbb{R}} x \, d\mu_3.$$

(**Hint:** To show the first equality (and the first part of the problem), start with a simple non-negative function h and follow the usual steps to conclude about general h. The second part of the problem is related to the first part.)

Answer

We utilize the following results:

- Proposition 2.1.1(ii): if T_1 is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$-measurable and T_2 is $\langle \mathcal{F}_2, \mathcal{F}_3 \rangle$-measurable, then $T_2 \circ T_1$ is $\langle \mathcal{F}_1, \mathcal{F}_3 \rangle$-measurable.
- Proposition 2.1.2: a continuous $f : \mathbb{R}^p \to \mathbb{R}^k$ is $\langle \mathcal{B}(\mathbb{R}^p), \mathcal{B}(\mathbb{R}^k) \rangle$-measurable.
- Proposition 2.2.1 and (2.2) (induced measure): if $T : \Omega_1 \to \Omega_2$ is $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$-measurable and μ is a measure on $(\Omega_1, \mathcal{F}_1)$, then $\mu T^{-1}(A) \equiv \mu(T^{-1}(A))$, $A \in \mathcal{F}_2$, is a measure on $\mathcal{F}_2$.
- Definitions 2.3.2 and 2.3.4 (integrals of simple nonnegative and of measurable functions; see Q1).
- Definition 2.3.3 (integral of a nonnegative measurable function): $\int f \, d\mu \equiv \lim_{n\to\infty} \int f_n \, d\mu$ for any nonnegative simple $f_n \uparrow f$. The limit does not depend on the sequence (Proposition 2.3.2), and such a sequence always exists, e.g. (3.7).
- Remark 2.3.7: f^+ and f^- are nonnegative and measurable, with $f = f^+ - f^-$ and $|f| = f^+ + f^-$.
- Definition 2.3.5 (integrability): $f \in L^1(\mu)$ iff $\int |f| \, d\mu < \infty$.

Proof. To keep the three spaces straight, note:

$$(\Omega_1, \mathcal{F}_1, \mu_1) \xrightarrow{f} (\Omega_2, \mathcal{F}_2, \mu_2) \xrightarrow{h} (\mathbb{R}, \mathcal{B}(\mathbb{R}), \mu_3), \qquad \mu_2 = \mu_1 f^{-1}, \quad \mu_3 = \mu_2 h^{-1}.$$

By Proposition 2.2.1, μ_2 and μ_3 are measures on $\mathcal{F}_2$ and $\mathcal{B}(\mathbb{R})$. By Proposition 2.1.1(ii), $g = h \circ f$ is $\langle \mathcal{F}_1, \mathcal{B}(\mathbb{R}) \rangle$-measurable. By Proposition 2.1.2, I is $\langle \mathcal{B}(\mathbb{R}), \mathcal{B}(\mathbb{R}) \rangle$-measurable, being continuous.

We first show

$$g \in L^1(\mu_1) \iff h \in L^1(\mu_2), \quad \text{in which case} \quad \int_{\Omega_1} g \, d\mu_1 = \int_{\Omega_2} h \, d\mu_2. \tag{Q4}$$

Following the hint, we build up from simple h.

Nonnegative simple h: Let $h = \sum_{i=1}^k c_i I_{A_i}$ with $c_i \in [0, \infty)$ and $A_i \in \mathcal{F}_2$. For $\omega \in \Omega_1$, $I_{A_i}(f(\omega)) = 1$ iff $f(\omega) \in A_i$ iff $\omega \in f^{-1}(A_i)$, so $I_{A_i} \circ f = I_{f^{-1}(A_i)}$. Thus

$$h \circ f = \sum_{i=1}^k c_i I_{f^{-1}(A_i)},$$

which is nonnegative and simple on Ω_1, since each $f^{-1}(A_i) \in \mathcal{F}_1$. By Definition 2.3.2 and (2.2),

$$\int_{\Omega_1} h \circ f \, d\mu_1 = \sum_{i=1}^{k} c_i \, \mu_1\big(f^{-1}(A_i)\big) = \sum_{i=1}^{k} c_i \, \mu_2(A_i) = \int_{\Omega_2} h \, d\mu_2.$$

Nonnegative measurable h: Let $h \geq 0$, and let $\{h_n\}_{n\geq 1}$ be nonnegative simple functions on Ω_2 with $h_n \uparrow h$ pointwise, e.g. as in (3.7). By the simple case, each $h_n \circ f$ is nonnegative and simple on Ω_1, and $h_n \circ f \uparrow h \circ f$ pointwise. Applying Definition 2.3.3 on both spaces, together with the simple case,

$$\int_{\Omega_1} h \circ f \, d\mu_1 = \lim_{n\to\infty} \int_{\Omega_1} h_n \circ f \, d\mu_1 = \lim_{n\to\infty} \int_{\Omega_2} h_n \, d\mu_2 = \int_{\Omega_2} h \, d\mu_2.$$

General h: Pointwise, $g^+ = h^+ \circ f$, $g^- = h^- \circ f$, and $|g| = |h| \circ f$. Since h^+, h^-, and $|h|$ are nonnegative and measurable (Remark 2.3.7), the previous case gives

$$\int_{\Omega_1} |g| \, d\mu_1 = \int_{\Omega_2} |h| \, d\mu_2 \qquad \text{and} \qquad \int_{\Omega_1} g^{\pm} \, d\mu_1 = \int_{\Omega_2} h^{\pm} \, d\mu_2.$$

By Definition 2.3.5, the first identity gives $g \in L^1(\mu_1)$ iff $h \in L^1(\mu_2)$. When both are integrable, the second identity and Definition 2.3.4 give

$$\int_{\Omega_1} g \, d\mu_1 = \int_{\Omega_1} g^+ \, d\mu_1 - \int_{\Omega_1} g^- \, d\mu_1 = \int_{\Omega_2} h^+ \, d\mu_2 - \int_{\Omega_2} h^- \, d\mu_2 = \int_{\Omega_2} h \, d\mu_2,$$

which proves (Q4).

Second equivalence: This is the same argument as for (Q4), one step down the chain: take $(\Omega_2, \mathcal{F}_2, \mu_2)$ as the starting space, h as the map, and I as the function being integrated. Since $I \circ h = h$ and $\mu_2 h^{-1} = \mu_3$, we show

$$h \in L^1(\mu_2) \iff I \in L^1(\mu_3), \quad \text{in which case} \quad \int_{\Omega_2} h \, d\mu_2 = \int_{\mathbb{R}} I \, d\mu_3.$$

Since I is not nonnegative, we work with a generic $\phi : \mathbb{R} \to \mathbb{R}$ and later take $\phi \in \{I^+, I^-, |I|\}$.

Nonnegative simple ϕ: Let $\phi = \sum_{i=1}^{k} c_i I_{B_i}$ with $c_i \in [0, \infty)$ and $B_i \in \mathcal{B}(\mathbb{R})$. For $\omega \in \Omega_2$, $I_{B_i}(h(\omega)) = 1$ iff $h(\omega) \in B_i$ iff $\omega \in h^{-1}(B_i)$, so $I_{B_i} \circ h = I_{h^{-1}(B_i)}$. Thus

$$\phi \circ h = \sum_{i=1}^{k} c_i I_{h^{-1}(B_i)},$$

which is nonnegative and simple on Ω_2, since each $h^{-1}(B_i) \in \mathcal{F}_2$. By Definition 2.3.2 and (2.2),

$$\int_{\Omega_2} \phi \circ h \, d\mu_2 = \sum_{i=1}^{k} c_i \, \mu_2\big(h^{-1}(B_i)\big) = \sum_{i=1}^{k} c_i \, \mu_3(B_i) = \int_{\mathbb{R}} \phi \, d\mu_3.$$

Nonnegative measurable ϕ: Let $\phi \geq 0$ be measurable, and let $\{\phi_n\}_{n\geq 1}$ be nonnegative simple functions on $\mathbb{R}$ with $\phi_n \uparrow \phi$ pointwise, e.g. as in (3.7). By the simple case, each $\phi_n \circ h$ is nonnegative and simple on Ω_2, and $\phi_n \circ h \uparrow \phi \circ h$ pointwise. Applying Definition 2.3.3 on both spaces, together with the simple case,

$$\int_{\Omega_2} \phi \circ h \, d\mu_2 = \lim_{n\to\infty} \int_{\Omega_2} \phi_n \circ h \, d\mu_2 = \lim_{n\to\infty} \int_{\mathbb{R}} \phi_n \, d\mu_3 = \int_{\mathbb{R}} \phi \, d\mu_3.$$

The identity I: Pointwise, $I^+ \circ h = h^+$, $I^- \circ h = h^-$, and $|I| \circ h = |h|$. Since I^+, I^-, and $|I|$ are nonnegative and measurable (Remark 2.3.7), the previous case gives

$$\int_{\Omega_2} |h| \, d\mu_2 = \int_{\mathbb{R}} |I| \, d\mu_3 \qquad \text{and} \qquad \int_{\Omega_2} h^{\pm} \, d\mu_2 = \int_{\mathbb{R}} I^{\pm} \, d\mu_3.$$

By Definition 2.3.5, the first identity gives $h \in L^1(\mu_2)$ iff $I \in L^1(\mu_3)$. When both are integrable, the second identity and Definition 2.3.4 give

$$\int_{\Omega_2} h\, d\mu_2 = \int_{\Omega_2} h^+\, d\mu_2 - \int_{\Omega_2} h^-\, d\mu_2 = \int_{\mathbb{R}} I^+\, d\mu_3 - \int_{\mathbb{R}} I^-\, d\mu_3 = \int_{\mathbb{R}} I\, d\mu_3 = \int_{\mathbb{R}} x\, d\mu_3.$$

Combining the two equivalences, $g \in L^1(\mu_1)$ iff $h \in L^1(\mu_2)$ iff $I \in L^1(\mu_3)$, and in that case $\int_{\Omega_1} g\, d\mu_1 = \int_{\Omega_2} h\, d\mu_2 = \int_{\mathbb{R}} x\, d\mu_3$. //

Q5: Extra A

Let $(\Omega, \mathcal{F}, P) = ([0,1], \mathcal{B}([0,1]), m)$, where m denotes the Lebesgue measure. Define the random variables $X, \{X_n\}$: for each $\omega \in \Omega$, $X(\omega) = 1$,

$$X_1(\omega) = 0, \qquad X_n(\omega) = 1 + (n-1) I_{[0, \frac{1}{n}]}(\omega), \quad n \geq 2.$$

We utilize the following results for (a)–(e):

- Proposition A.2.1 (Axiom of Archimedes; see Q1).
- Definition 2.3.2 (integral of a simple nonnegative function; see Q1).
- Definition 2.5.2 (convergence a.e.; see Q1). Convergence a.s. (P) is convergence a.e. with respect to P.
- Definitions 2.5.3 and 2.5.4 (convergence in measure and in L^p; see Q3). Convergence in probability is convergence in measure with respect to P.
- Theorem 2.5.1: if $\mu(\Omega) < \infty$, then $f_n \to f$ a.e. (μ) implies $f_n \xrightarrow{m} f$.
- Definition 2.5.8 (uniform integrability; see Q1).

(a)

Does $X_n \to X$ everywhere as $n \to \infty$?

Answer

Narp; the single point $\omega = 0$ spoils the fun.

Proof. For all $n \geq 2$, $0 \in [0, \frac{1}{n}]$, so

$$X_n(0) = 1 + (n-1) = n \to \infty \neq 1 = X(0). \quad //$$

(b)

Does $X_n \to X$ a.s. (P) as $n \to \infty$?

Answer

Yes (yarp).

Proof. Let $\omega \in (0,1]$ be given. By Proposition A.2.1, there exists $N \in \mathbb{N}$ with $N > \max\{1/\omega, 1\}$, so $N \geq 2$. For all $n \geq N$, $1/n \leq 1/N < \omega$, so $\omega \notin [0, \frac{1}{n}]$ and $X_n(\omega) = 1 = X(\omega)$. So $X_n(\omega) \to X(\omega)$ for every $\omega \in (0,1]$. Together with (a),

$$B \equiv \{\omega \in [0,1] : X_n(\omega) \not\to X(\omega)\} = \{0\}.$$

Since $P(\{0\}) = m(\{0\}) = 0$, $X_n \to X$ a.s. (P) (Definition 2.5.2). //

(c)

Does $X_n \to X$ in probability as $n \to \infty$?

Answer

Yes (yarp, again).

Proof. $P = m$ is a probability measure on $[0,1]$, so $P(\Omega) = 1 < \infty$. By (b), $X_n \to X$ a.s. (P), so Theorem 2.5.1 gives $X_n \xrightarrow{P} X$. //

(d)

Does $X_n \to X$ in L^p as $n \to \infty$? Argue for different values of $p \in (0, \infty)$.

Answer

Narp (but also yarp); $X_n \to X$ in L^p for $0 < p < 1$, but not for $1 \le p < \infty$.

Proof. First, X and every X_n are in $L^p(P)$ for every $0 < p < \infty$: $|X| = 1$, $X_1 = 0$, and $|X_n| \le n$ for $n \ge 2$, so each is bounded, and $P(\Omega) = 1$. Next, for $n \ge 2$, $X_n - X = (n-1)I_{[0,\frac{1}{n}]}$, so $|X_n - X|^p = (n-1)^p I_{[0,\frac{1}{n}]}$ is nonnegative and simple. By Definition 2.3.2,

$$\int |X_n - X|^p \, dP = (n-1)^p \, m\big([0, \tfrac{1}{n}]\big) = \frac{(n-1)^p}{n}.$$

So, by Definition 2.5.4, $X_n \to X$ in L^p iff $(n-1)^p/n \to 0$.

Case $0 < p < 1$*:* $0 \le (n-1)^p/n < n^p/n = n^{p-1} \to 0$, since $p - 1 < 0$. So $X_n \to X$ in L^p.

Case $p = 1$*:* $(n-1)/n = 1 - 1/n \to 1 \ne 0$. So $X_n \not\to X$ in L^1.

Case $1 < p < \infty$*:* Since $(n-1)/n \ge 1/2$ for $n \ge 2$,

$$\frac{(n-1)^p}{n} = (n-1)^{p-1} \cdot \frac{n-1}{n} \ge \frac{(n-1)^{p-1}}{2} \to \infty,$$

since $p - 1 > 0$. So $X_n \not\to X$ in L^p. //

(e)

Is $\{X_n\}$ a uniformly integrable sequence of random variables?

Answer

Narp.

Proof. Let $a_n(t)$ be as in Q1, with P in place of μ, and let $t \ge 1$ be given. Take any $n > t$, so that $n \ge 2$. On $[0, \frac{1}{n}]$, $|X_n| = n > t$, and on $(\frac{1}{n}, 1]$, $|X_n| = 1 \le t$. So $\{|X_n| > t\} = [0, \frac{1}{n}]$, and by Definition 2.3.2,

$$a_n(t) = \int n I_{[0,\frac{1}{n}]} \, dP = n \, m\big([0, \tfrac{1}{n}]\big) = 1.$$

Thus $\sup_{n \ge 1} a_n(t) \ge 1$ for every $t \ge 1$, which does not tend to 0 as $t \to \infty$. Hence $\{X_n\}_{n \ge 1}$ is not UI (Definition 2.5.8). //